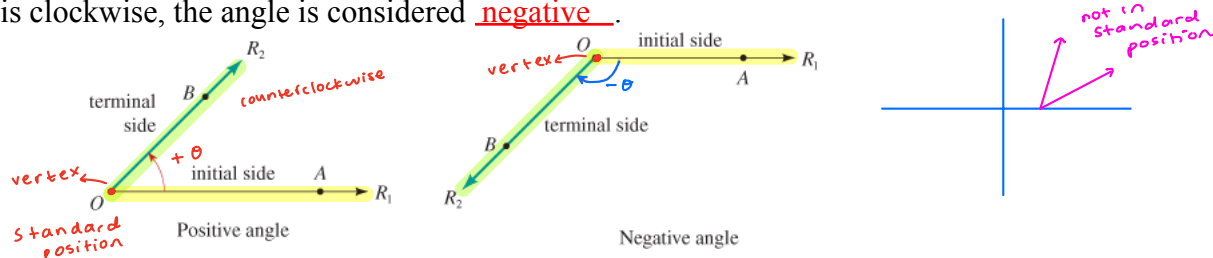


An **angle** consists of two rays with a common vertex. We often interpret an angle as a rotation of the ray R_1 onto R_2 . In this case, R_1 is called the initial side, and R_2 is called the terminal side of the angle. If the rotation is counterclockwise, the angle is considered **positive**, and if the rotation is clockwise, the angle is considered negative.

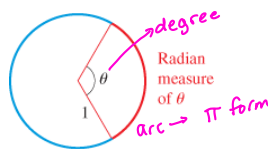


I. Angle Measure

The **measure** of an angle is the amount of rotation about the vertex required to move R_1 onto R_2 . One unit of measurement for angles is the **degree**. An angle of measure 1 degree is formed by rotating the initial side $1/360$ of a complete revolution.

Definition of Radian Measure: If a circle of radius 1 is drawn with the vertex of an angle at its center, then the measure of this angle in **radians** (abbreviated rad) is the length of the arc

that subtends the angle.

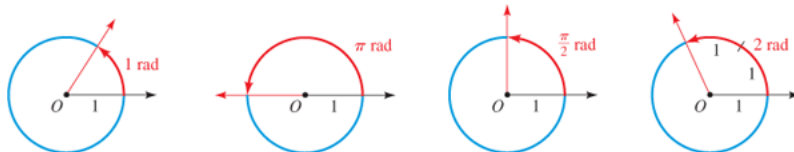


For a circle of radius 1, a complete revolution has measure 2π rad; a straight angle has measure π rad; and a right angle has measure $\pi/2$ rad.

Relationship Between Degrees and Radians

To convert degrees to radians, multiply by $\pi/180$. To convert radians to degrees, multiply by $180/\pi$.

$$180^\circ = \pi \text{ rad} \quad 1 \text{ rad} = \left(\frac{180}{\pi}\right)^\circ \quad 1^\circ = \frac{\pi}{180} \text{ rad}$$



degrees → radians

$$\times \frac{\pi}{180}$$

$$120^\circ \rightarrow 120 \cdot \frac{\pi}{180} = \frac{2\pi}{3}$$

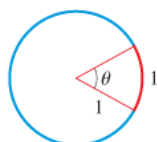
radians → degrees

$$\times \frac{180}{\pi}$$

$$\frac{5\pi}{4} \rightarrow \frac{5\pi}{4} \cdot \frac{180}{\pi} = 225^\circ$$

$$1 \text{ rad} \approx 57.296^\circ \quad \text{and} \quad 1^\circ \approx 0.01745 \text{ rad}$$

$$1 \cdot \frac{180}{\pi}$$



Measure of $\theta = 1 \text{ rad}$
Measure of $\theta \approx 57.296^\circ$

When no unit is given, the angle is assumed to be measured in radians.

Example 1: Express

a) 1000° in radians

$$= 1000 \cdot \frac{\pi}{180}$$

$$= \boxed{\frac{50\pi}{9}}$$

b) -3.5 rad in degrees

$$= -3.5 \cdot \frac{180}{\pi}$$

$$\approx \boxed{-200.54^\circ}$$

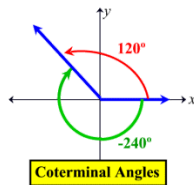
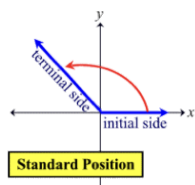
c) $\frac{5\pi}{6}$ rad in degrees.

$$= \frac{5\pi}{6} \cdot \frac{180}{\pi}$$

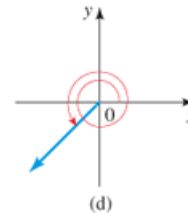
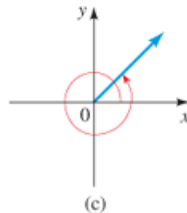
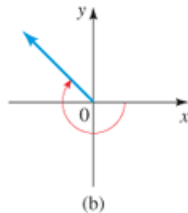
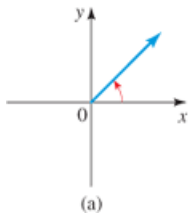
$$= \boxed{150^\circ}$$

II. Angles in Standard Position

An angle is in **standard position** if it is drawn in the xy -plane with its vertex at the origin and its initial side on the positive x -axis. Two angles in standard position are **coterminal** if their sides coincide.



coterminal angles: $\boxed{\theta \pm 360^\circ n}$



Example 2: Coterminal Angles

A. The measure of an angle in standard position is given. Find two positive angles and two negative angles that are coterminal with the given angle.

i. 135°

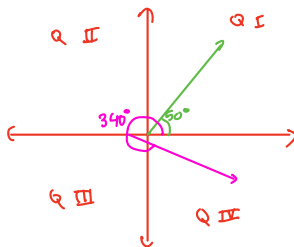
$$\begin{aligned} 135^\circ + 360^\circ &= \boxed{495^\circ} \\ 135^\circ + 720^\circ &= \boxed{855^\circ} \\ 135^\circ - 360^\circ &= \boxed{-225^\circ} \\ 135^\circ - 720^\circ &= \boxed{-585^\circ} \end{aligned}$$

ii. $\frac{11\pi}{6}$

$$\begin{aligned} \frac{11\pi}{6} + 2\pi &= \boxed{\frac{23\pi}{6}} & \frac{11\pi}{6} + 4\pi &= \boxed{\frac{35\pi}{6}} \\ \frac{11\pi}{6} - 2\pi &= \boxed{-\frac{\pi}{6}} & \frac{11\pi}{6} - 4\pi &= \boxed{-\frac{13\pi}{6}} \end{aligned}$$

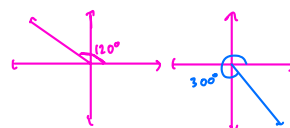
B. The measures of two angles in standard position are given. Determine whether the angles are coterminal.

i. $50^\circ, 340^\circ$



No

ii. $\frac{32\pi}{3}, \frac{11\pi}{3}$



No

$$\begin{aligned} \frac{32(180^\circ)}{3} &= \frac{1920^\circ}{3} = 640^\circ \\ \frac{11(180^\circ)}{3} &= \frac{660^\circ}{3} = 220^\circ \end{aligned}$$

III. Length of a Circular Arc

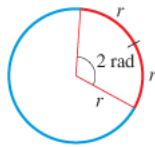
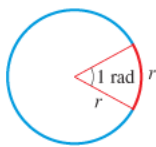
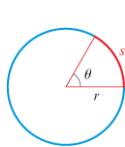
An angle whose radian measure is θ is subtended by an arc that is the fraction $\frac{\theta}{2\pi}$ of the circumference of a circle. Hence, in a circle of radius r , the length s of an arc that subtends a central angle of θ radians

$$s = \frac{\theta}{2\pi} \times \text{circumference of circle} = \frac{\theta}{2\pi} (2\pi r) = r\theta, \quad \boxed{s = r\theta} \quad \text{or, solving for } \theta, \text{ we get } \underline{\underline{\theta = s/r}}$$

This formula allows us to define radian measure using a circle of radius r : The radian measure of an angle θ is the number of radiuses that can fit in the arc that subtends θ .

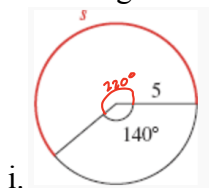
$$s = r\theta, \quad \theta = \frac{s}{r}, \quad r = \frac{s}{\theta}$$

↙ angle ↘ ↙ radius ↘

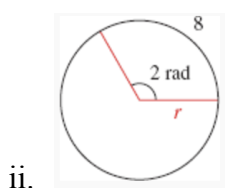


Example 3:

- a. Find the length s of the circular arc, the radius r of the circle, or the central angle, as indicated.

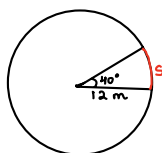


$$\begin{aligned} s &= r\theta \\ &= 5 \left(220^\circ \cdot \frac{\pi}{180^\circ} \right) \\ &= \boxed{\frac{55}{9}\pi} \end{aligned}$$



$$r = \frac{s}{\theta} = \frac{8}{2} = \boxed{4}$$

- b. Find the length s of the arc that subtends a central angle of 40° measure in a circle of radius 12 m.



$$\begin{aligned} s &= r\theta \\ &= 12 \left(40^\circ \cdot \frac{\pi}{180^\circ} \right) \\ &= \boxed{\frac{8\pi}{3} \text{ m}} \end{aligned}$$

- c. A central angle θ in a circle of radius 9 m is subtended by an arc of length 14 m. Find the measure of θ in degrees and radians.

$$s = 14 \text{ m}$$

$$\theta = \frac{s}{r} = \boxed{\frac{14}{9}}$$

$$\theta = \frac{14}{9} \cdot \frac{180^\circ}{\pi} \approx \boxed{89.12^\circ}$$

- d. Find the radius r of the circle if an arc of length 15 m on the circle subtends a central angle of $\frac{5\pi}{6}$.

$$\begin{aligned} s &= r\theta \\ r &= \frac{s}{\theta} \end{aligned}$$

$$r = \frac{15}{\frac{5\pi}{6}}$$

$$= \frac{15 \cdot 6}{5\pi}$$

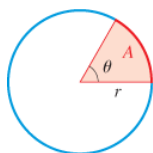
$$= \frac{18}{\pi}$$

$$r \approx \boxed{5.73 \text{ m}}$$

IV. Area of a Circular Sector

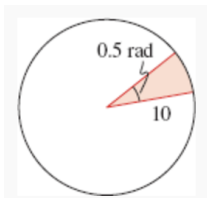
The area of a circle of radius r is $A = \pi r^2$. A sector of this circle with central angle θ has an area that is the fraction $\frac{\theta}{2\pi}$ of the area of the entire circle. Therefore, in a circle of radius r , the area A of a sector with a central angle of θ radians is

$$A = \frac{\theta}{2\pi} (\text{area of circle}) = \frac{\theta}{2\pi} (\pi r^2) = \boxed{\frac{1}{2} r^2 \theta} \quad \boxed{A = \frac{1}{2} r^2 \theta} \rightarrow \text{area of sector}$$



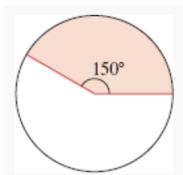
Example 4:

- a. Find the area of the sector shown in each figure.



$$\begin{aligned}
 A &= \frac{1}{2} r^2 \theta \\
 &= \frac{1}{2} (10^2) (0.5) \\
 &= \boxed{25 \text{ unit}^2}
 \end{aligned}$$

- b. Find the radius of each circle if the area of the sector is 12.



$$\begin{aligned}
 A &= \frac{1}{2} r^2 \theta \\
 12 &= \frac{1}{2} r^2 \left(150 \cdot \frac{\pi}{180} \right) \\
 24 &= r^2 \left(\frac{5\pi}{6} \right) \\
 r &= \sqrt{\frac{24}{\frac{5\pi}{6}}} \\
 r &= \sqrt{\frac{144}{5\pi}} \\
 r &\approx \boxed{3.03 \text{ units}}
 \end{aligned}$$

- c. The area of a circle is 600 m^2 . Find the area of a sector of this circle that subtends a central angle of 3 rad.

$$\begin{aligned}
 A &= \pi r^2 \\
 600 &= \pi r^2 \\
 \frac{600}{\pi} &= r^2
 \end{aligned}$$

$$\begin{aligned}
 A &= \frac{1}{2} r^2 \theta \\
 &= \frac{1}{2} \left(\frac{600}{\pi} \right) (3) \\
 &\approx \boxed{286.48 \text{ m}^2}
 \end{aligned}$$

- d. A sector of a circle of radius 80 mi has an area of 1600 mi^2 . Find the central angle (in radians) of the sector.

$$\begin{aligned}
 r &= 80 \\
 A &= 1600 \\
 A &= \frac{1}{2} r^2 \theta \\
 1600 &= \frac{1}{2} (80^2) \theta \\
 \theta &= \boxed{0.5 \text{ rad}}
 \end{aligned}$$

V. Circular Motion

There are two ways to describe the motion of a point on a circle.

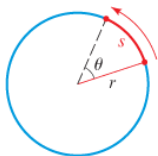
Linear speed is the rate at which the distance traveled is changing, so linear speed is the distance traveled divided by the time elapsed. **Angular speed** is the rate at which the central angle θ is changing, so angular speed is the number of radians this angle changes divided by the time elapsed.

Suppose a point moves along a circle of radius r and the ray from the center of the circle to the point traverses θ radians in time t . Let $s = r\theta$ be the distance the point travels in time t . Then the speed of the object is given by

Angular speed: $\omega = \frac{\theta}{t}$

Linear speed: $v = \frac{s}{t}$

$$|v| = \frac{s}{t}$$



(The symbol ω is the Greek letter “omega.”)

Relationship Between Linear and Angular Speed

If a point moves along a circle of radius r with angular speed ω , then its linear speed v is given by

$$v = r\omega \quad . \quad v = \frac{s}{t} = \frac{r\theta}{t} = r\omega$$

Example 5: A car's wheels are 28 in. in diameter. How far (in miles) will the car travel if its wheels revolve 10,000 times without slipping?

$$d = 28 \text{ in}, \quad r = 14 \text{ in}$$
$$10,000 \cdot 28\pi \text{ in} \cdot \frac{1 \text{ ft}}{12 \text{ in}} \cdot \frac{1 \text{ mi}}{5280 \text{ ft}} \approx 13.88 \text{ mi}$$

Example 6: How many revolutions will a car wheel of diameter 30 in. make as the car travels a distance of one mile?

$$d = 30 \text{ in}, \quad r = 15 \text{ in}$$
$$1 \text{ mi} \cdot \frac{5280 \text{ ft}}{1 \text{ mi}} \cdot \frac{12 \text{ in}}{1 \text{ ft}} = 63360 \text{ in}$$
$$C = \pi \cdot 30 \text{ in} = 30\pi \text{ in}$$
$$\frac{63360 \text{ in}}{30\pi \text{ in}} \approx 672.27 \text{ revolutions}$$

Example 7: A ceiling fan with 16-inch blades rotates at 45 rpm.

(a) Find the angular speed of the fan in rad/min.

$$\omega = \frac{45}{1} \frac{\text{rev}}{\text{min}} \cdot \frac{2\pi \text{ rad}}{1 \text{ rev}} = 90\pi \text{ rad/min}$$

(b) Find the linear speed of the tips of the blades in in./min.

angular speed calculated above: $90\pi \text{ rad/min}$

$$v = r\omega$$
$$v = 16 \text{ in} \cdot \frac{90\pi \text{ rad}}{1 \text{ min}}$$
$$= \frac{1440\pi \text{ in}}{\text{min}}$$
$$\approx 4523.89 \text{ in/min}$$